

Do LLM Self-Explanations Tell One Story?

A Chance-, Ceiling-, and Reliability-Corrected Audit of Cross-Paradigm Agreement

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Abstract

Large language models are routinely asked to explain their own predictions through interchangeable-looking devices: token highlighting, free-text rationales, counterfactual rewrites, and importance rankings. We ask whether these elicitation paradigms actually name the same evidence, in a pre-registered study of 3 model families (two open-weight, one proprietary) $\times$ 4 datasets (2,400 instances), including the shortcut-resistant CAD-IMDb. Because raw token overlap is confounded by chance and by set-size geometry, we introduce ECS_{adj} : an exact-hypergeometric chance correction combined with a geometric ceiling correction and paradigm-balanced aggregation; and because the elicitation instruments are themselves unreliable, we measure per-strategy self-consistency ceilings under three independent prompt rewordings and apply classical reliability disattenuation. Agreement exceeds chance in every model $\times$ dataset cell (12/12, Holm $p \leq 0.0012$; pooled $ECS_{adj} = 0.43$ on a 0=chance, 1=ceiling scale), but is far from uniform: after reliability correction, extraction $\leftrightarrow$ counterfactual divergence largely disappears (corrected agreement 0.88–0.94), whereas rationale-involving divergence persists (rationale–counterfactual 0.39, 95% CI [0.35, 0.43]). Two models explaining the same input with the same paradigm agree more than one model explaining it with two paradigms, on all four datasets. Consensus tokens are causally load-bearing—erasing them flips predictions 2.0–2.7 $\times$ more than matched random controls and 4–5 $\times$ more than size-matched *non-consensus* salient tokens—yet a pre-registered test finds agreement does *not* predict simulatability under explanation-independent perturbations. Explanation paradigms are not interchangeable views of one underlying story, and internal agreement cannot be assumed to certify downstream utility.

1 Introduction

When practitioners want to know *why* an LLM made a prediction, they ask it—and the asking can take several standard forms: score the importance of each token (Huang et al., 2023), give a one-sentence rationale (Camburu et al., 2018; Wiegrefe et al., 2021), produce a minimal edit that would flip the label (Ross et al., 2021), or rank the most important words. These paradigms are used interchangeably in practice, as if they were different renderings of one underlying explanation. For post-hoc explainers of classical models, the “disagreement problem” (Krishna et al., 2022) showed that different attribution methods routinely name different evidence. Whether the same fragmentation afflicts the *self*-explanations of instruction-tuned LLMs—one model, one prediction, four ways of asking—is the question of this paper.

Answering it is mostly a measurement problem, with three obstacles. First, *chance*: two random token subsets of a short input overlap substantially, so raw Jaccard rewards brevity. Second, *geometry*: a counterfactual edit touches 1–2 tokens while a salience list contains 3–10, capping achievable Jaccard far below 1 for some pairs but not others, so a flat average measures set-size geometry more than agreement. Third, *reliability*: the instruments are themselves noisy—under mild prompt paraphrase, per-strategy self-agreement ranges from 0.62 to 0.90—so a low observed value is ambiguous between *real paradigm divergence* and *unreliable elicitation*.

We address all three. Our estimator, ECS_{adj} , normalizes each pairwise Jaccard by its exact hypergeometric expectation (chance floor) and its size-determined maximum (ceiling), then aggregates the five cross-paradigm pairs into three paradigm-level contrasts with equal weight. Per-strategy test–retest reliabilities, measured with a

paraphrase ablation on every model, feed a classical Spearman disattenuation (Spearman, 1904): if the instruments were perfectly reliable, how much would the paradigms agree? All estimands, test families, exclusion floors, and amendments were pre-registered before collection—including a fourth, counterfactually-revised dataset (Kaushik et al., 2020) that blunts the “near-solved benchmark” objection, and a simulatability bridge (Chen et al., 2024) whose result we report as the null it produced.

Findings. On 2,400 instances (3 models $\times$ 4 datasets $\times$ 200), elicited at temperature 0 through frozen, literature-anchored prompts:

1. **Above-chance agreement everywhere.** Complete-case ECS_{adj} is positive in all 12 model $\times$ dataset cells (range 0.28–0.62; Holm-corrected sign-flip $p \leq 0.0012$ in every cell), pooled 0.43—roughly 40% of the way from chance to the geometric ceiling (§5.1).
2. **Agreement is structured by paradigm distance, and reliability correction sharpens the structure.** Observed agreement orders extraction–perturbation $>$ extraction–rationalization $>$ rationalization–perturbation (complete-case within-instance means $0.615 > 0.395 > 0.286$). After disattenuation, extraction $\leftrightarrow$ counterfactual pairs approach the ceiling (H–CF 0.88, RO–CF 0.96)—their observed divergence is largely elicitation noise—while every rationale-involving pair stays far below 1 (R–CF 0.39, CI [0.35, 0.43]): free-text rationales genuinely select different evidence (§5.2).
3. **Explanations are paradigm-shaped more than model-shaped.** Two *different* models explaining the same input with the same strategy agree more than *one* model explaining it with two strategies, on all 4 datasets (matched paired contrast, all CIs above 0; §5.3).
4. **Consensus is causally load-bearing but not decision-useful.** Erasing the tokens that ≥ 3 strategies agree on flips the model’s own prediction $2.0\text{--}2.7\times$ more often than occurrence-matched random controls, and $4\text{--}5\times$ more than equally-sized non-consensus salient tokens (all models, both operators, $p_{Holm}=0.0002$; §5.4). Yet agreement does not predict simulatability under explanation-independent perturbations: explanations barely improve a simulator over a

no-explanation baseline (pooled gains -0.015 to $+0.016$), and the pre-registered association between ECS_{adj} and simulatability gain is null (§5.5).

We release the full pipeline: frozen prompts with hashes, curated datasets with datasheets, per-instance artifacts, seeds, and the pre-registration with dated amendments.

2 Related Work

Faithfulness and self-consistency. Interpretability work distinguishes an explanation being *plausible* from being *faithful* to the model’s computation (Jacovi and Goldberg, 2020; Lyu et al., 2024). Turpin et al. (2023) showed chain-of-thought can misstate the true drivers of predictions, and Madsen et al. (2024) found self-explanation faithfulness is model- and task-dependent. Parcalabescu and Frank (2024) argue most existing tests—including erasure-style checks—measure *self-consistency* rather than faithfulness. We adopt that framing: all our axes are consistency measures, and none certifies faithfulness (§7).

Agreement between explanation methods. The disagreement problem (Krishna et al., 2022) quantified how post-hoc attribution methods conflict for classical models; for LLMs, Huang et al. (2023) compared self-generated saliency to LIME/occlusion and Randl et al. (2024) report related single-model reliability results. We differ in three ways: we compare four *self*-elicitation paradigms (not model-external attributions) across three model families and four datasets; we correct for both chance and set-size ceiling with an exact null; and we read observed agreement against measured instrument reliability—to our knowledge the first application of psychometric disattenuation to explanation agreement.

Counterfactuals and simulatability. Minimal contrastive editing (Ross et al., 2021) defines the counterfactual construct we elicit, and Dehghanighobadi et al. (2025) study LLM self-counterfactuals in isolation. Kommiya Mothilal et al. (2021) formalize why attribution (sufficiency-oriented) and counterfactual (necessity-oriented) evidence diverge by construction; our rationalization–perturbation results quantify that divergence once chance, ceiling, and reliability are controlled. Counterfactual

simulatability (Chen et al., 2024) operationalizes an explanation’s practical value—does it help an observer predict model behavior on perturbed inputs?—and we pre-registered a bridge from agreement to a strategy-fair variant of it (§5.5). For verbalized confidence we replicate the over-confidence clustering of Xiong et al. (2024) and Tian et al. (2023).

3 Measuring Cross-Paradigm Agreement

3.1 Four elicitation paradigms, one token space

For each instance the model first classifies the text; every explanation is then elicited for the model’s own predicted label (a label-conditioned, post-hoc protocol; see §7 for what this implies). The four strategies are: **H** (highlighting): score every word 1–10 for importance (Huang et al., 2023); the evidence set is a length-proportional top- k , with repeated words aggregated by max score and score ties broken alphabetically in the normalized token space for determinism. The rule as registered and as implemented is $k = \max(3, \text{round}(L/5))$, where L counts the input’s whitespace tokens that are not pure punctuation, capped at the number of valid scored tokens. It adapts Huang et al. (2023)’s length-proportional selection with two disclosed deviations: a floor of 3 rather than 1, and rounding rather than the floor function (their §III-B prints $\min(1, \lfloor L/5 \rfloor)$, which yields $k \leq 1$ for every input and contradicts their own top-3 prompt; the intended reading is max). Because k is a design choice with no canonical value, we also report a size-robust composite built from overlap coefficients rather than Jaccard—a post-hoc recombination of the plan’s pre-registered overlap secondary—which preserves the paradigm ordering (E–P 0.786 > E–R 0.689 > R–P 0.526 pooled); a fully k -free graded-weight variant is left to future work. **R** (rationale): a one-sentence free-text explanation (Wiegrefe et al., 2021); content words are extracted with a POS filter. **CF** (counterfactual): a minimal edit that flips the label, constrained to $\leq \frac{1}{3}$ of words (Ross et al., 2021); the evidence set is the tokens changed, and validity requires the model’s verified label flip. **RO** (rank ordering): the 5 most important words, ranked. Three paradigms group these four strategies: *extraction* (E) = {H, RO}, *rationalization* (R) = {R}, *perturbation* (P) = {CF}. In pair names (H–R) letters denote strategies; in compo-

nent names (E–R, R–P) they denote paradigms. All evidence sets are mapped into one shared token space (lowercasing, punctuation stripping, stopword removal, lemmatization), so “terrible” and “Terrible.” match across strategies. The H–RO pair is same-paradigm and serves as a within-paradigm reference; it is excluded from the cross-paradigm composite.

3.2 The ECS_{adj} estimator

For evidence sets A, B drawn from an instance vocabulary of size V , raw Jaccard J has two structural confounds. Under uniform random selection the intersection is hypergeometric, giving the *exact* chance expectation

$$\mathbb{E}[J] = \sum_k \binom{a}{k} \binom{V-a}{b-k} \binom{V}{b}^{-1} \frac{k}{a+b-k}, \quad (1)$$

with $a=|A|, b=|B|$ (no Monte-Carlo error, no seed). And the best achievable value is the nesting ceiling $J_{\max} = \min(a, b) / \max(a, b)$: a *perfect* 1-token CF against a 5-token salience set caps at $J = 0.2$. We therefore score every pair with the adjusted Jaccard

$$AJ(A, B) = \frac{J - \mathbb{E}[J]}{J_{\max} - \mathbb{E}[J]}, \quad (2)$$

a κ -style normalization: 0 = chance, 1 = maximum agreement achievable given the sizes, comparable across pairs, instances, and datasets regardless of geometry. Pairs with $J_{\max} - \mathbb{E}[J] < \varepsilon$ (pre-registered $\varepsilon = 0.10$) are degenerate—chance and ceiling nearly coincide—and are excluded with a counted flag rather than allowed to dominate the mean. AJ is not bounded below by -1 : its floor is $(J_{\min} - \mathbb{E}[J]) / (J_{\max} - \mathbb{E}[J])$, which equals $-\mathbb{E}[J] / (J_{\max} - \mathbb{E}[J])$ in the usual case but is higher when $a + b > V$ forces a non-empty intersection ($J_{\min} = (a + b - V) / V$); 5.8% of scored pairs are in this forced-overlap regime, and the ε guard does not exclude them. The exact null already integrates over this constraint (the hypergeometric sum starts at $k = \max(0, a + b - V)$), so the estimand is unaffected—only the floor is instance-dependent. Because the resulting null is left-skewed (bounded above at +1, long negative tail), a one-sided sign-flip test for mean > 0 rejects less often than a symmetric-null test would at the same effect size: the permutation distribution inherits the same skew, so extreme negative draws inflate its upper tail and raise the rejection

threshold. Negative cell means are therefore reported only as “not above chance,” never as signed below-chance effects.

The five cross-paradigm pairs aggregate into three paradigm contrasts with equal weight,

$$\begin{aligned} \text{E-R} &= \frac{1}{2}[\text{AJ}(\text{H}, \text{R}) + \text{AJ}(\text{RO}, \text{R})], \\ \text{E-P} &= \frac{1}{2}[\text{AJ}(\text{H}, \text{CF}) + \text{AJ}(\text{RO}, \text{CF})], \\ \text{R-P} &= \text{AJ}(\text{R}, \text{CF}), \end{aligned}$$

and $\text{ECS}_{\text{adj}} = \text{mean}(\text{E-R}, \text{E-P}, \text{R-P})$. This paradigm balancing reduces CF’s share of the composite from 3/5 to 1/3—important because CF is by far the least reliable strategy. The *primary estimand* is complete-case ECS_{adj} (all three components defined); the available-component variant is a labelled sensitivity, never the headline, because for over half its rows it reflects a single paradigm pair.

3.3 Reliability ceilings and disattenuation

Each elicitation strategy is an *instrument*, and instruments have test–retest reliability. We measure it per model $\times$ dataset $\times$ strategy by re-eliciting every explanation with semantically equivalent paraphrases of the prompt on a seeded 50-instance cohort, scoring same-strategy agreement on the AJ scale: $\text{rel}_s = \text{AJ}(\text{base}_s, \text{paraphrase}_s)$. We use *three* independent rewordings of each of the four elicitation prompts, so every ceiling is an average over rewordings rather than a property of one; §5.2 reports how far the corrected values move when each rewording is used alone. Pooled reliabilities are R 0.89, RO 0.79, H 0.65, CF 0.62 (per-cell values in Appendix Table 11); even at temperature 0 with byte-identical prompts, no model exactly reproduced its own salience list in a single-instance retest probe (Jaccard 0.82–0.91), consistent with batched-inference nondeterminism (Thinking Machines Lab, 2025).

Observed between-instrument agreement is attenuated by instrument unreliability; the classical correction (Spearman, 1904) is

$$\text{corr}(A, B) = \text{obs}(A, B) / \sqrt{\text{rel}_A \cdot \text{rel}_B}. \quad (3)$$

The interpretation was registered in advance: corrected ≈ 1 means the pair’s divergence is within elicitation noise; corrected $\ll 1$ with a CI excluding 1 means real paradigm divergence beyond instrument unreliability. Spearman’s formula is defined for product-moment correlations, so applying it to mean agreement levels is an analogue,

which we validate with planted-agreement simulations on the AJ scale (Appendix C): disattenuation cuts mean recovery error roughly fourfold, is essentially unbiased when both reliabilities are ≥ 0.60 (bias +0.01)—the regime of every pooled pair below—and acquires an upward bias only near the 0.30 floor, which is what the floor exists to exclude. Pairs are estimable only if both reliabilities ≥ 0.30 with ceiling $n \geq 10$; excluded pairs are reported as excluded (in this run, the three CF pairs of one cell, Nova Pro/AG News, where CF paraphrase reliability falls to 0.10). One population caveat belongs here. The ceiling admits a CF elicitation whenever it *parses* into a non-empty edit set, whereas the main analysis admits CF only on a *verified* label flip. The gate is therefore weaker in the denominator than in the numerator, which is why cells with very low CF validity still clear $n \geq 10$ (Qwen3-235B/AG News: CF validity 7.5%, but ceiling $n = 13$ —the smallest in the study; a validity-gated cohort would have yielded ≈ 4). Because the extra elicitations are ones whose edit failed to flip, they are plausibly noisier, so rel_{CF} is if anything *under*-estimated—which inflates corrected CF pairs. This pushes in the same direction as the at-ceiling caveat in §5.2 and is a further reason to read those values as upper bounds. Values above 1 (sampling error) are reported with an at-ceiling flag, never truncated. CIs come from a seeded cluster bootstrap ($B=2,000$) resampling instances and ceiling draws independently.

3.4 Pre-registration and inference

The analysis plan was written and amended in dated stages, all *before* the production run: the estimator and its validation gates; the estimand–test alignment; the per-dataset co-primary granularity and instrument fixes; and a final amendment adding the reliability correction, the CAD-IMDb arm, and the simulatability bridge. Family (a), primary: per-cell one-sided sign-flip permutation tests (10,000 permutations, seed 42) of complete-case $\text{ECS}_{\text{adj}} > 0$, Holm-corrected (Holm, 1979) across the 12 cells. Family (a2): the same test on available components (sensitivity). Family (b): consensus-core erasure vs. matched random control, Holm across operators. Family (c): the simulatability bridge (§5.5). Everything else is descriptive. The registered plan with its dated amendments is available to reviewers as an anonymized

read-only record,¹ and the per-instance artifacts are released with the code.

4 Experimental Setup

Models. Amazon Nova Pro, Qwen3-235B-A22B-2507, and DeepSeek-V3—two open-weight families (Qwen3, DeepSeek) and one proprietary, API-only model (Nova Pro), all served through AWS Bedrock (temperature 0, fixed decoding parameters). The trio reflects availability constraints, not a principled sample of frontier systems (§7).

Datasets. SST-2 (Socher et al., 2013), MNLI (Williams et al., 2018), AG News (Zhang et al., 2015), and CAD-IMDb—the counterfactually *revised* IMDb reviews of Kaushik et al. (2020), in which human editors minimally rewrote reviews to flip the label, breaking single-cue shortcuts. Each dataset is a frozen, hand-vetted, label-balanced 200-instance set curated from the source split (dedup, length stratification, quality filters), with datasheets recording provenance and licenses. CAD-IMDb predates model training cut-offs; it addresses the shortcut prong of the easy-benchmark objection, not memorization (§7).

Pipeline. $3 \times 4 \times 200 = 2,400$ instances; per instance: classification, verbalized confidence, and the four explanations (26,545 API requests, 7.1M tokens; 100% instance completion; refusals affected 35 of 2,400 instances, 1.5%). Model accuracy spans 80–98% per cell; explanations are elicited for the model’s own prediction regardless of correctness. Strategy-level validity is H 98%, R 97%, RO 99%, and CF 59% (range 8–99% per cell)—CF’s minimal-edit constraint frequently fails to produce a verified flip, itself a finding replicating the validity–minimality trade-off: unconstrained “free” rewrites flip reliably (90%) but over-edit (0.32 vs. 0.11 edited-token ratio). Complete cases (all three components defined) number 1,153 of 2,400; missingness is CF-driven and handled by the dual complete/available reporting plus a free-CF sensitivity re-analysis (§5.6).

Erasure protocol. For each instance we erase the consensus core CC3 (tokens named by ≥ 3 of 4 strategies; present in 92% of instances, mean size

2.8) by masking or deletion, re-classify with the *same* model, and compare flip rates against a random control matched on content-word type *and* occurrence counts. CF rewrites are additionally verified by a held-out model (96% cross-model flip confirmation).

Simulatability protocol. On a seeded 50-instance/dataset/model cohort (600 rows), we generate three explanation-independent perturbations per instance (two seeded random content-word erasures, one deterministic top-frequency deletion), obtain the target model’s actual labels on them, then ask a *different* model (round-robin simulator) to predict those labels given the original text, the prediction, and *one* strategy’s explanation, versus a no-explanation baseline. $\text{gain}_s = \text{aCC}_s - \text{aCC}_{\text{baseline}}$.

5 Results

5.1 Agreement is above chance in every cell

Table 1 shows the primary result. Complete-case ECS_{adj} is positive in all 12 cells, from 0.284 (Nova Pro/MNLI) to 0.617 (DeepSeek-V3/SST-2), and every cell survives the Holm-corrected sign-flip test at $p_{\text{Holm}} = .0012$. The pooled mean of 0.432 places the four ways of asking “why” about 40% of the way from chance toward the maximum achievable given each pair’s evidence-set sizes. The available-component family (a2) agrees (12/12, all $p_{\text{Holm}} = .0012$). The cell means conceal wide, often bimodal per-instance spread—instances with perfect nesting ($\text{AJ} = 1$) coexist with instances at or below chance in every cell (Figure 2, appendix).

The complete-case and available-component columns diverge most on MNLI (0.28–0.44 vs. 0.48–0.51), and the reason is mechanical rather than substantive. MNLI has the second-lowest counterfactual validity (36–46%), so most available-component rows there lose their CF-bearing components and reduce to E–R alone—the component that sits well above R–P (0.53 vs. 0.31 pooled on this dataset). The available estimate is thus higher precisely *because* it drops the hardest paradigm contrast, which is why we pre-registered the complete-case population as primary and report the wider- N variant only beside it. Where CF validity is high (CAD-IMDb, $\geq 92\%$) the two columns nearly coincide, as expected under this account.

¹ANONYMIZED PRE-REGISTRATION: [OSF view-only link--insert before submission]. Timestamps for each amendment are visible in the registration history.

Dataset	Model	Complete-case (primary)			Available (sensitivity)	
		N	ECS_{adj}	p_{Holm}	N	ECS_{adj}
SST-2	Nova Pro	109	0.414	.0012	186	0.426
SST-2	Qwen3-235B	125	0.449	.0012	188	0.443
SST-2	DeepSeek-V3	143	0.617	.0012	197	0.618
MNLI	Nova Pro	37	0.284	.0012	182	0.475
MNLI	Qwen3-235B	39	0.431	.0012	191	0.508
MNLI	DeepSeek-V3	51	0.441	.0012	193	0.501
AG News	Nova Pro	30	0.355	.0012	190	0.439
AG News	Qwen3-235B	14	0.427	.0012	195	0.489
AG News	DeepSeek-V3	80	0.503	.0012	194	0.544
CAD-IMDb	Nova Pro	154	0.301	.0012	198	0.287
CAD-IMDb	Qwen3-235B	178	0.370	.0012	199	0.354
CAD-IMDb	DeepSeek-V3	193	0.465	.0012	200	0.459
Pooled		1153	0.432	—	2313	0.462

Table 1: **Family (a), primary result.** Complete-case ECS_{adj} (0 = chance, 1 = size-adjusted ceiling) per model×dataset cell, with the pre-registered one-sided sign-flip test (Holm across 12 cells; all raw $p \leq 2 \times 10^{-4}$). The available-component estimate is the pre-registered wider- N sensitivity family (a2, also 12/12 significant, all $p_{Holm}=.0012$). Pooled rows are descriptive (instances cluster within models). Complete-case N varies because completeness requires a verified minimal counterfactual; see §5.6 for the missingness analysis.

Pair	Comp.	Obs.	$rel_A \times rel_B$	Corrected [95% CI]
RO-CF	E-P	0.653	$.79 \times .62$	0.936 [0.898, 0.978]
H-CF	E-P	0.555	$.65 \times .62$	0.876 [0.829, 0.922]
H-R	E-R	0.495	$.65 \times .89$	0.649 [0.621, 0.676]
RO-R	E-R	0.424	$.79 \times .89$	0.504 [0.481, 0.527]
R-CF	R-P	0.291	$.89 \times .62$	0.393 [0.354, 0.432]

Table 2: **Observed vs. reliability-corrected pairwise agreement**, pooled; the same data as Figure 1 (per-cell values and exclusions in Appendix Table 11). Obs. = mean AJ where the pair is defined; rel = each strategy’s self-consistency ceiling, pooled over *three* independent prompt rewodings; Corrected = Spearman-disattenuated. Every corrected CI excludes 1. The same-paradigm H-RO reference (observed 0.596) is excluded from ECS_{adj} and carries no registered correction.

Model and dataset structure is systematic: DeepSeek-V3 is the most self-consistent model in all four datasets (model-pooled 0.52 vs. Qwen3-235B 0.41, Nova Pro 0.34), and the shortcut-resistant CAD-IMDb is the hardest dataset for every model (dataset-pooled 0.38)—when single lexical cues are broken by design, the paradigms drift further apart, exactly where explanations matter most.

5.2 Paradigm distance, and what reliability correction reveals

Agreement is not flat across paradigm pairs (Table 2, Figure 1). Observed agreement follows a ladder ordered by paradigm distance—within-instance complete-case component means: E-P

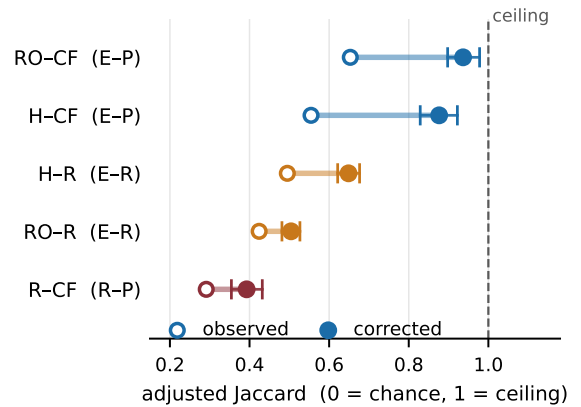


Figure 1: Observed (hollow) vs. reliability-corrected (filled) agreement for the five cross-paradigm pairs, pooled; whiskers are bootstrap CIs on the corrected value. Correcting for instrument unreliability moves the two extraction-perturbation pairs to the ceiling but leaves every rationale-involving pair far short of it.

0.615 > E-R 0.395 > R-P 0.286 ($N=1,153$)—with the same-paradigm H-RO reference at 0.596. Remarkably, the extraction-counterfactual pairs agree about as strongly as the two extraction methods agree with each other, *before* any reliability correction.

The disattenuation resolves the ambiguity that a raw ladder leaves open. Correcting each pair by its instruments’ self-consistency ceilings, the two extraction-perturbation pairs rise to 0.88 and 0.94—close to, though statistically just short of, the ceiling (both CIs exclude 1). In other words, *most of the observed extraction↔counterfactual*

divergence is elicitation noise, not paradigm disagreement: when a minimal edit exists and is found, it touches the same tokens the extractive paradigms score highest. By contrast, every rationale-involving pair stays far below its ceiling: H–R 0.649, RO–R 0.504, and R–CF 0.393—even though the rationale is the *most* reliable instrument ($\text{rel} = 0.89$), so its divergence cannot be blamed on noisy elicitation. The one-sentence free-text rationale systematically names different evidence than both the extractive and the counterfactual paradigms. A pre-registered soft-matching sensitivity (§5.6) shows this is not lexical variation (synonyms explain $\approx 0\%$ of the gap): rationales differ *evidentially*.

Because a low reliability *inflates* the corrected ratio, this reading depends on the ceilings being properties of the instruments rather than of one particular rewording. We therefore collected the ceilings under *three* independent paraphrases of every elicitation prompt, across all 12 cells (§3.3). Individual cells do move—Nova Pro’s SST-2 CF reliability ranges 0.57–0.70 across the three wordings—but pooled over cells the corrected values are stable: re-deriving Table 2 under each paraphrase separately shifts every pair by at most 0.043 (H–CF 0.853–0.895; RO–CF 0.923–0.956; R–CF 0.391–0.396, a range of 0.005), and the ordering never changes. Both extraction–perturbation pairs land just below the ceiling under all three. The remaining exposure is that the recovery simulation models *independent* instrument noise, whereas the shared label anchor plausibly correlates errors across paradigms in the direction that inflates corrected values (Appendix C); and per-cell entries whose CF reliability nears the floor (e.g. DeepSeek-V3/AG News) still plausibly overshoot and carry at-ceiling flags for that reason. The rationale-divergence conclusion is exposed to none of this: R is the *most* reliable instrument, the most paraphrase-stable one, and its corrected pairs sit 0.35–0.61 below the ceiling.

The pattern is strongest exactly where short-cuts are broken: on CAD-IMDb, corrected R–CF falls to 0.12–0.36 across models while five of the six corrected extraction–CF pairs lie between 0.90 and 1.04 (the exception is Nova Pro’s H–CF at 0.53; Appendix Table 11). This is consistent with the necessity/sufficiency distinction of Kommiya Mothilal et al. (2021): attribution-style and counterfactual evidence coincide when one cue is both sufficient and necessary, but free-text ratio-

Dataset	Cross	Within	Δ_{matched} [95% CI]
SST-2	0.622	0.498	+0.129 [0.073, 0.190]
MNLI	0.596	0.495	+0.046 [0.009, 0.087]
AG News	0.569	0.491	+0.066 [0.035, 0.100]
CAD-IMDb	0.585	0.367	+0.218 [0.187, 0.248]

Table 3: **Cross-model contrast** (AJ scale). Cross = same strategy, different models, mean AJ; Within = one model’s cross-paradigm ECS_{adj} . Δ_{matched} is the pre-registered strategy-matched paired per-instance contrast ($\{\text{H}, \text{R}, \text{RO}\}$ cross-model pairs vs. the within-model extraction–rationalization component), which controls CF representation; all four CIs exclude 0.

nalization drifts toward plausible narrative rather than the tokens that actually decide the flip.

5.3 Explanations are paradigm-shaped, not model-shaped

If each explanation paradigm were a window onto a model-specific reasoning process, one model’s four explanations should cohere more than two different models’ renditions of the same paradigm. The opposite holds (Table 3): same-strategy agreement *across* models exceeds cross-paradigm agreement *within* a model on all four datasets, with the strategy-matched paired contrast positive everywhere (from +0.046 on MNLI to +0.218 on CAD-IMDb). The elicitation paradigm, not model identity, is the dominant factor shaping which evidence an explanation names. Practically: swapping the model changes the explanation less than swapping the question format.

Cross-model pairs are *not* conditioned on the two models having predicted the same label: they agree on 93.9% of the 8,074 pairs, and the remaining 6.1% compare explanations of different predicted labels. This makes the contrast conservative—restricting to label-matched pairs could only raise the cross-model side—so the direction of the finding is unaffected.

5.4 Consensus cores are causally load-bearing

Do the tokens the paradigms agree on actually matter to the prediction? Erasing the CC3 consensus core flips the model’s own prediction in 28–33% of instances, versus 10–15% for occurrence-matched random controls—a $2.0\text{--}2.7\times$ ratio, significant for every model under both erasure operators (Table 4; Figure 3 in the appendix plots both panels). The gap grows monotonically with agreement tier (pooled, delete operator: 0.152 $\rightarrow$

Model	Op.	CC3	Random	Gap	p_{Holm}
DeepSeek-V3	mask	0.316	0.123	0.192	.0002
DeepSeek-V3	delete	0.330	0.125	0.205	.0002
Nova Pro	mask	0.291	0.147	0.144	.0002
Nova Pro	delete	0.311	0.132	0.179	.0002
Qwen3-235B	mask	0.276	0.104	0.172	.0002
Qwen3-235B	delete	0.305	0.112	0.193	.0002

Table 4: **Family (b): consensus-core erasure.** Flip rate of the model’s own prediction after erasing CC3 tokens (named by ≥ 3 strategies) vs. an occurrence-matched random content-word control; one-sided sign-flip permutation on within-instance paired differences, Holm across operators.

Baseline	Op.	Baseline	CC3	$\Delta [p_{\text{Holm}}=]$
SS1	mask	0.064	0.289	+0.224 [.0002]
SS1	delete	0.058	0.305	+0.247 [.0002]
TF-IDF	mask	0.074	0.294	+0.219 [.0002]
TF-IDF	delete	0.081	0.314	+0.233 [.0002]
random (Table 4)		0.125	0.294	+0.170

Table 5: **Consensus vs. non-consensus salience** (*post-hoc*). Erasing size- and occurrence-matched *non-consensus* tokens—SS1 (named by exactly one strategy) or top-TF-IDF (excluding the core)—flips the model’s prediction far less often than erasing the consensus core. Δ is the paired within-instance difference (one-sided sign-flip, Holm across operators within each baseline; $n=2,037\text{--}2,179$). Both baselines fall *below* the random control, which does not exclude consensus tokens and so hits them by chance.

0.206 $\rightarrow$ 0.213 for low/mid/high-ECS_{adj} terciles; descriptive). Following Parcalabescu and Frank (2024), we interpret this strictly as a second *consistency* axis—stated evidence coincides with revealed input sensitivity—not as ground-truth faithfulness: the core is extracted from, and the flip verified by, the same model (Atanaseva et al., 2020; DeYoung et al., 2020).

Consensus vs. salience (*post-hoc*). A matched-random control shows consensus beats *noise*; it does not show consensus beats any salient-token heuristic. We therefore ran two stricter baselines that erase *non-consensus* tokens matched to CC3 on both type count and destroyed-occurrence count, re-classifying with each instance’s own model: **SS1**, tokens named by exactly one strategy (model-selected, but no consensus formed), and **TF-IDF**, the instance’s top-TF-IDF content tokens excluding the core (lexical salience that never consults the model). Both are far less disruptive than the consensus core (Table 5): pooled flip rates are

0.06 (SS1) and 0.07–0.08 (TF-IDF) against CC3’s 0.29–0.31, a 4–5 $\times$ ratio, with the paired within-instance difference significant for every model under both operators ($p_{\text{Holm}}=0.0002$). Restricting to instances where the baseline reached CC3’s full erasure budget (SS1 65%, TF-IDF 96% of instances) leaves the gap essentially unchanged (+0.185 to +0.226).

Notably both salience baselines flip *less* than the random control (0.12). This is not paradoxical: the random control samples the instance’s content words without excluding consensus tokens, so it lands on them by chance, whereas these arms exclude the core by construction. The random control is thus the weaker comparison, and the effect against a genuinely non-consensus baseline is roughly twice as large as the headline number. A size-normalized view agrees: per erased token, CC3 flips at 0.096 (mask) and 0.103 (delete), above every individual strategy—CF 0.072/0.079, RO 0.072/0.078, R 0.057/0.061, H 0.035/0.036—whose whole evidence sets flip more often only because they destroy far more text (mean 12.6 tokens for H, 6.2 for CF, against 3.1 for CC3). Consensus tokens are the densest evidence per unit of text destroyed, and that density does not come from salience alone.

5.5 Agreement does not predict simulatability under generic perturbations

Cross-paradigm agreement would be decision-relevant if it flagged when explanations fail their core job: helping an observer predict model behavior (Chen et al., 2024). One construct note: Chen et al. (2024) derive probe inputs from the explanation under evaluation, whereas our perturbations are deliberately explanation-*independent* so that one perturbation set is fair to all four strategies. We therefore test whether explanations help predict behavior on *generic* corruptions—a weaker claim than explanation-derived simulatability, and why we qualify this subsection’s conclusions accordingly. The pre-registered test answers no (Table 6). First, a stark descriptive fact: explanations barely help at all. Simulator accuracy with an explanation differs from the no-explanation baseline by -0.07 to $+0.04$; the largest gain (DeepSeek-V3 rationales, $+0.038$) is small, and counterfactuals *hurt* simulation of Qwen3-235B (-0.069). Second, the registered association is null: instance-level ECS_{adj} does not

Target model	Baseline acc.	Simulatability gain over baseline				$\rho(\text{ECS}_{\text{adj}}, \text{gain})$ [95% CI]	p_{Holm}
		H	R	CF	RO		
Nova Pro	0.877	−0.009	−0.002	−0.006	−0.001	−0.011 [−0.155, 0.144]	1.00
Qwen3-235B	0.870	+0.005	+0.012	−0.069	+0.003	+0.006 [−0.141, 0.172]	1.00
DeepSeek-V3	0.767	+0.016	+0.038	+0.020	+0.013	−0.015 [−0.159, 0.146]	1.00
Pooled	0.838	+0.004	+0.016	−0.015	+0.005	−0.001 [−0.091, 0.088]	—

Table 6: **Family (c), pre-registered null.** Simulatability on 600 cohort instances under *explanation-independent* perturbations (a deliberately strategy-fair variant of the explanation-derived construct of [Chen et al., 2024](#); see text): a held-out simulator model predicts the target model’s labels on perturbed inputs, given one strategy’s explanation vs. no explanation. Explanations barely move simulator accuracy, and instance-level ECS_{adj} does not predict simulatability gain (c1 Spearman shown; the c2 tercile contrasts are likewise null, all Holm $p \geq 0.99$).

correlate with simulatability gain for any model (c1: $|\rho| \leq 0.015$, all Holm $p=1.0$; c2 tercile contrasts: all Holm $p \geq 0.99$), and a low-agreement “red flag” has precision ≤ 0.04 for identifying instances where explanations mislead. High baselines are part of the story (perturbed predictions are guessable from the text alone at 77–88%, leaving limited headroom), but the conclusion within this design is unambiguous: internal consistency among self-explanations is not a proxy for their utility to an observer *on this task family*. Whether agreement predicts the stronger, explanation-derived simulatability construct remains open; testing it requires per-strategy perturbation sets and is the natural follow-up. We report this null as pre-registered; it constrains how our—and others’—agreement metrics should be used.

5.6 Robustness

Six pre-registered sensitivity analyses (i–vi) and two post-hoc checks added in revision (vii–viii) probe the main results. **(i) Missingness:** complete-case membership is CF-gated (a verified minimal edit must exist). Substituting the unconstrained free-CF rewrite for the minimal one yields complete-case ECS_{adj} 0.382 ($N=1,318$), positive in all cells: conclusions survive removing the minimal-edit gate. **(ii) Chance model:** the uniform null assumes evidence tokens are drawn uniformly from the instance vocabulary, whereas all strategies preferentially select salient-looking words; a frequency-weighted null shifts the pooled estimate by only -0.011 ($0.432 \rightarrow 0.421$), with every cell still positive. The shift is small by construction: the urn is the *content-word* vocabulary after stopword removal and lemmatization, so the high-frequency function words that would otherwise dominate a frequency-weighted draw are already absent, leaving a compara-

tively flat weight distribution over the remaining candidates. **(iii) Lexical variation:** scoring with embedding-based soft matching (pinned static vectors, cosine $\tau=0.8$, matched MC null) leaves complete-case ECS_{adj} unchanged (0.4320 at $\tau=0.8$) and closes $\approx 0\%$ of the extraction–perturbation vs. rationale-pair gap ($+0.200$ hard vs. $+0.204$ soft): the rationale divergence of §5.2 is evidential, not synonymy. **(iv) Estimand family:** the available-component family (a2) reproduces 12/12 significance. **(v) Test conservatism:** the AJ floor asymmetry makes the one-sided sign-flip test conservative by construction. **(vi) Confidence:** verbalized confidence is degenerate at temperature 0 (mean 0.94 with heavy clustering at 0.95), replicating [Xiong et al. \(2024\)](#); its association with ECS_{adj} is weak and inconsistent (per-cell Spearman -0.16 to $+0.31$; Appendix Table 9) and we draw no conclusion from it. **(vii) Degeneracy exclusions (post-hoc):** the ε guard drops pairs unevenly (17.3% of H–R and 17.0% of H–CF slots vs. 2.5% of RO–R), so we re-derived the ladder on the 850 complete cases with *no* degenerate pair: E–P $0.618 > \text{E–R}$ $0.451 > \text{R–P}$ 0.313 —same ordering as the full set ($0.615/0.395/0.286$), so the guard does not manufacture it (Appendix E). **(viii) Prediction correctness (post-hoc):** complete-case ECS_{adj} is 0.437 on correct predictions ($n=1,085$) vs. 0.360 on incorrect ones ($n=68$), with no consistent per-cell direction (Table 10): agreement is not an artifact of correct instances, though the incorrect stratum is too small for inference.

6 Discussion

Paradigms are complements, not substitutes. The reliability-corrected picture says the four standard ways of asking “why” collapse into roughly two evidential objects, not one and not four: an

extraction/counterfactual cluster agreeing near its noise ceiling, and free-text rationalization, which reliably (in both senses) tells a different story. This cautions against averaging paradigms into one “explanation quality” score, and against treating a rationale as a cheap summary of token-level evidence—on shortcut-broken data the two are barely related (corrected R-P as low as 0.12).

Measure the instrument before the construct. Uncorrected, our data would “show” substantial extraction–counterfactual disagreement (observed 0.56–0.65); about three quarters of that apparent shortfall is elicitation noise, which the paraphrase ceiling makes visible and the disattenuation removes. Agreement studies of LLM explanations should routinely report self-consistency ceilings—observed agreement without them conflates two very different failure modes.

Consistency $\neq$ utility. The erasure result says consensus evidence is causally load-bearing; the simulatability null says none of this measurably helps an observer predict the model on generic corruptions of the input. Both can be true: our erasure perturbs the tokens explanations name, while simulatability requires explanations to transfer *decision structure*, which one-shot label-conditioned explanations of high-accuracy classifiers apparently do not carry—at least under explanation-independent perturbations. Agreement metrics—ours included—should therefore be read as descriptions of the explanation-generating process, not as certificates of usefulness.

7 Conclusion

Four self-explanation paradigms, measured across three LLM families and four datasets under a pre-registered design, agree above chance everywhere—but not uniformly. Once instrument reliability is accounted for, counterfactual and extractive evidence coincide up to elicitation noise while free-text rationales diverge evidentially; the paradigm shapes the explanation more than the model does; consensus tokens are causally load-bearing; and none of this predicts simulatability under explanation-independent perturbations. The measurement stack (exact-null adjusted Jaccard, paradigm balancing, paraphrase ceilings, disattenuation) is model-agnostic and, we hope, useful wherever explanation agreement is quantified.

Limitations

Consistency, not faithfulness. Every axis here—cross-paradigm agreement, erasure, simulatability—is a consistency measure among model outputs (Parcalabescu and Frank, 2024; Jacovi and Goldberg, 2020). High agreement does not certify that any explanation reflects the model’s computation; all four paradigms could converge on the same plausible-but-unfaithful cue.

Label-conditioned elicitation. All explanation prompts reveal the model’s predicted label, so we measure consistency of *post-hoc rationalizations* (Wiegrefe et al., 2021; Turpin et al., 2023). The shared anchor could inflate agreement by pointing every strategy at the same obvious cue; the chance/ceiling adjustment does not remove this, and a no-label control is future work.

Counterfactual-driven selection. Complete-case ECS_{adj} conditions on a verified minimal counterfactual existing; 1,153/2,400 instances (48%) qualify. Complete and incomplete instances are indistinguishable on every agreement component measurable for both, on length, vocabulary, and accuracy, and the free-CF sensitivity reproduces all conclusions, but selection on unobservables cannot be excluded. Relatedly, the attribution $\leftrightarrow$ intervention gap we quantify is partly definitional (Kommiya Mothilal et al., 2021): flip-sets and support-sets need not coincide; our contribution is measuring the size and reliability-adjusted reality of that gap, not adjudicating it as error.

Models and data. Three mid-to-large model families—two open-weight, one proprietary—accessed through one provider; none is a current frontier system (an access constraint, not a design choice). All four datasets predate training cut-offs and are plausibly contaminated (Sainz et al., 2023); accuracy is 80–98%, so we describe explanations of largely easy predictions—CAD-IMDb blunts the shortcut concern but not the contamination concern. English classification tasks only; generation and reasoning tasks are untested.

Instrument scope. Reliability ceilings rest on three paraphrases per prompt over 50-instance cohorts. Three is enough to show that the pooled corrected values are not artifacts of one rewording (spread ≤ 0.043 ; §5.2) but not enough to

characterize the full distribution of paraphrases, and individual cells remain sensitive: a cell’s reliability for one strategy moves by up to 0.26 across wordings (DeepSeek-V3/MNLI, H), so per-cell corrected values should be read with wider uncertainty than their bootstrap CIs alone suggest. The ceilings are also paraphrase-stability estimates specifically, not full test–retest distributions — they hold the model, decoding, and cohort fixed and vary only the wording. A distinct and unresolved exposure is that our validation of the correction (Appendix C) models instrument noise as *independent* across strategies, whereas the shared predicted-label anchor plausibly correlates errors across paradigms. Correlated errors inflate observed agreement without inflating self-consistency, so they inflate the corrected ratio too—in the direction of the at-ceiling claim. That caveat does not touch the rationale-divergence result (§5.2), which is a claim that corrected agreement is *low*. Verbalized confidence is degenerate at temperature 0 and uninformative here. Simulatability used three perturbations per instance and one simulator per target; low power against very small effects cannot be excluded, though the point estimates are near-exactly zero.

Ethics Statement

All data are public English-language benchmark texts; no human subjects, annotators, or personal data were involved. The study consumed approximately 7.1M tokens of commercial LLM inference. Findings indicate that LLM self-explanations, while internally consistent above chance, are paradigm-dependent and of unproven utility to observers; deploying them as trust signals in high-stakes settings would be premature, and our results should not be cited as evidence that any studied model “explains itself” faithfully.

Reproducibility Statement

The full pipeline (collection, parsing, normalization, metrics, statistics, figures), the frozen prompt templates with SHA-256 manifest, curated dataset files with datasheets, the dated pre-registration with all amendments, per-instance artifacts, and every number in this paper are generated from a single archived run (seed 42, temperature 0, checkpointed and resumable; 600+ unit/property tests, including planted-agreement simulations validating the estimator and, added in

revision, the disattenuation step). Analyses are labelled pre-registered or post-hoc at each point of use. Code and artifacts will be released upon publication.

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A Elicitation Prompts

All strategies are elicited with frozen templates (SHA-256 manifest in the release); the classification and confidence prompts, and dataset-specific CF variants (multiclass label sets, NLI hypothesis editing), appear in the release. The four core templates, abridged to their instruction lines ($\{\cdot\}$ = filled fields; every prompt opens with “The text was classified as: $\{predicted\ label\}$ ” and the text, and requests JSON-only output):

H (highlighting). “Analyze the importance of each word for this classification. Give every word an importance score from 1 (irrelevant) to 10 (decisive), using each word exactly as it appears in the text, in order. Include every occurrence, even repeated words.”

R (rationale). “In one sentence, explain why the text was classified as {predicted label}.”

CF (counterfactual). “Edit the text so that it is classified as {other label(s)} instead. Make as few edits as possible—change at most a third of the words, and do not add new sentences. Keep the original spacing and punctuation style; only change the words you edit.”

RO (rank ordering). “Identify the 5 most important words for this classification, ranked from most to least important. Each item must be a single word copied from the text.”

Each strategy also has a frozen semantically-equivalent paraphrase used only for the self-consistency ceiling (§3.3).

B Coverage, Components, and Confidence

Table 7 reports per-strategy validity and complete-case counts per cell. Table 8 reports the per-cell paradigm components (available-component means; the within-instance complete-case comparison is in §5.2). Table 9 reports the verbalized-confidence association, and Table 10 the complete-case estimate split by prediction correctness. Figure 4 shows the component heatmap beside the raw-Jaccard view it replaces; Figure 5 shows the counterfactual validity–minimality trade-off; Figure 6 the confidence–agreement grid.

C Validating the Disattenuation on the AJ Scale

Spearman’s correction is derived for product-moment correlations; we apply it to mean adjusted-Jaccard agreement, so we validate the analogue by simulation rather than assume it.

Design. Each instrument observes a latent evidence set $T_s \subseteq V$. One elicitation keeps each token of T_s independently with probability q_s and replaces each dropped token with a uniform random distractor—size-preserving, matching AJ’s fixed-size geometry. For a configuration $(V, a, b, \text{overlap}, q_A, q_B)$ we draw 2,000 elicitation pairs and compute rel_s as mean AJ between two independent elicitations of the same T_s , obs as mean AJ across instruments, and $\text{true} = \text{AJ}(T_A, T_B)$ noise-free. The grid crosses four geometries (balanced, imbalanced, large- V , small- V), $q \in \{0.55, 0.7, 0.85, 1.0\}$, and three planted

overlap levels: 120 configurations, 66 above the pre-registered floor.

Result. Above the floor, disattenuation recovers the planted value with mean $|\text{corrected} - \text{true}| = 0.050$, against mean $|\text{obs} - \text{true}| = 0.260$ on the noisy configurations—a fourfold error reduction. Bias depends on reliability: in the band $\text{min}(\text{rel}) \geq 0.60$ recovery is essentially unbiased (signed bias $+0.009$, max error 0.041), in $[0.45, 0.60]$ bias is $+0.047$, and in $[0.30, 0.45]$ it reaches $+0.072$ (max $+0.156$, on configurations with planted agreement 1.0). Below the floor, maximum error grows to 0.245. Two consequences for the paper: every pooled pair in Table 2 sits in the unbiased regime (lowest pooled reliability is CF at 0.62), and the upward near-floor bias means low-reliability *per-cell* entries should be read as upper bounds—which is what the at-ceiling flag marks. These assertions are pinned in the released test suite, in `tests/` as `test_disattenuation_recovery.py`.

Scope. The simulation draws each instrument’s noise independently. It therefore validates the correction’s algebra on the AJ scale but does *not* cover correlated instrument errors, which our design plausibly induces: every strategy is conditioned on the same predicted label (§Limitations), so the paradigms may err toward the same salient token together. Correlated errors raise observed agreement without raising same-strategy self-consistency, and so inflate the corrected ratio. The at-ceiling reading is the claim exposed to this; the rationale-divergence reading, being a claim of *low* corrected agreement, is if anything conservative under it.

D Paraphrase Spread of the Reliability Ceilings

Each of the four elicitation prompts has three independent rewordings (the originally registered paraphrase plus two added in revision). All preserve the format slots and the JSON response schema; only surface wording differs. Every ceiling in Table 2 pools all three.

Ceilings move with wording; pooled corrected values do not. Per cell the choice of rewording matters: Nova Pro’s SST-2 ceilings span 0.50–0.59 (H), 0.57–0.70 (CF) and 0.70–0.81 (RO) across the three, and the rationale instrument is the most stable (0.86–0.90). Pooled across the twelve

Table 7: Per-strategy extraction success and complete-case counts, per model $\times$ dataset.

Cell	N	H	R	CF	RO	Complete
DeepSeek-V3/AG News	200	1.000	0.960	0.415	1.000	81
DeepSeek-V3/CAD-IMDb	200	1.000	1.000	0.985	1.000	197
DeepSeek-V3/MNLI	200	1.000	0.965	0.460	0.965	88
DeepSeek-V3/SST-2	200	1.000	0.955	0.905	0.990	172
Nova Pro/AG News	200	0.995	0.950	0.155	1.000	31
Nova Pro/CAD-IMDb	200	0.965	0.985	0.920	0.995	177
Nova Pro/MNLI	200	0.995	0.940	0.355	0.970	65
Nova Pro/SST-2	200	0.990	0.960	0.650	0.990	124
Qwen3-235B/AG News	200	0.975	0.975	0.075	1.000	15
Qwen3-235B/CAD-IMDb	200	0.955	0.990	0.930	1.000	179
Qwen3-235B/MNLI	200	0.945	0.990	0.400	0.975	77
Qwen3-235B/SST-2	200	1.000	0.940	0.810	0.985	149

Table 8: Paradigm-component adjusted Jaccard: E-R, E-P, R-P per cell.

Cell	E-R	E-P	R-P
DeepSeek-V3/AG News	0.556	0.522	0.456
DeepSeek-V3/CAD-IMDb	0.403	0.704	0.277
DeepSeek-V3/MNLI	0.526	0.501	0.367
DeepSeek-V3/SST-2	0.522	0.770	0.581
Nova Pro/AG News	0.446	0.376	0.275
Nova Pro/CAD-IMDb	0.233	0.531	0.090
Nova Pro/MNLI	0.531	0.248	0.191
Nova Pro/SST-2	0.442	0.485	0.331
Qwen3-235B/AG News	0.490	0.444	0.410
Qwen3-235B/CAD-IMDb	0.251	0.703	0.138
Qwen3-235B/MNLI	0.543	0.477	0.338
Qwen3-235B/SST-2	0.339	0.750	0.258

Table 9: Confidence $\leftrightarrow$ ECS association (Spearman ρ , Kendall τ_b), per cell.

Cell	ρ	τ_b	p	N
DeepSeek-V3/AG News	-0.113	-0.090	0.1177	194
DeepSeek-V3/CAD-IMDb	0.310	0.235	0.0000	200
DeepSeek-V3/MNLI	0.026	0.020	0.7172	192
DeepSeek-V3/SST-2	0.215	0.169	0.0024	198
Nova Pro/AG News	-0.164	-0.130	0.0244	189
Nova Pro/CAD-IMDb	0.079	0.062	0.2702	197
Nova Pro/MNLI	0.207	0.156	0.0046	187
Nova Pro/SST-2	0.209	0.164	0.0034	194
Qwen3-235B/AG News	-0.009	-0.005	0.9065	190
Qwen3-235B/CAD-IMDb	0.146	0.112	0.0410	196
Qwen3-235B/MNLI	0.122	0.092	0.0973	186
Qwen3-235B/SST-2	0.175	0.137	0.0137	198

cells, however, that variation largely averages out. Re-deriving the whole overall table under each rewording *separately* gives:

Pair	wording 1	wording 2	wording 3	range
RO-CF	0.956	0.923	0.931	0.034
H-CF	0.883	0.895	0.853	0.043
H-R	0.654	0.666	0.628	0.038
RO-R	0.516	0.499	0.499	0.017
R-CF	0.391	0.396	0.392	0.005

No pair moves by more than 0.043, the ordering is identical under all three wordings, and the rationale-perturbation contrast is the most stable quantity in the study (0.005). The single-paraphrase design would have reported RO-CF as 0.956 with a CI that includes 1; pooling three rewordings gives 0.936 with a CI that excludes it, which is why we describe the extraction-perturbation pairs as approaching rather than reaching the ceiling.

E Degeneracy Exclusions by Pair Type

The $\varepsilon = 0.10$ guard removes pairs whose geometry puts the ceiling within ε of the chance

floor, i.e. $J_{\max} - \mathbb{E}[J] < \varepsilon$. Exclusion rates differ sharply by pair type: H-R 17.3% of elicited slots (396/2286), H-CF 17.0% (238/1397), R-CF 13.7% (189/1375), RO-CF 4.8% (67/1403), RO-R 2.5% (58/2304); RO’s fixed 5-token sets rarely produce degenerate geometry.

Two distinct regimes. The condition can be met from either side, and which one fires is dataset-dependent:

(a) **Small vocabulary.** On short inputs V is small, so $\mathbb{E}[J]$ is large and rises to meet an otherwise healthy ceiling. This is a genuinely short-input effect: it accounts for 65 of SST-2’s exclusions (median $V = 10$) and 41 of MNLI’s (median $V = 12$), where the degenerate pairs average $\mathbb{E}[J] = 0.37$ and 0.18 respectively.

(b) **Extreme size imbalance.** $J_{\max} = \min/\max$ is itself below ε regardless of V , because one evidence set is far larger than the other. Here chance is negligible—degenerate pairs average $\mathbb{E}[J] \approx 0.045$ —and the ceiling alone is the binding constraint.

Cell	Correct		Incorrect	
	ECS _{adj}	<i>N</i>	ECS _{adj}	<i>N</i>
DeepSeek-V3/AG News	0.505	70	0.483	10
DeepSeek-V3/CAD-IMDb	0.475	187	0.154	6
DeepSeek-V3/MNLI	0.463	40	0.360	11
DeepSeek-V3/SST-2	0.615	140	0.705	3
Nova Pro/AG News	0.350	27	0.398	3
Nova Pro/CAD-IMDb	0.306	145	0.222	9
Nova Pro/MNLI	0.297	33	0.171	4
Nova Pro/SST-2	0.406	102	0.530	7
Qwen3-235B/AG News	0.427	14	—	0
Qwen3-235B/CAD-IMDb	0.372	170	0.345	8
Qwen3-235B/MNLI	0.469	35	0.095	4
Qwen3-235B/SST-2	0.444	122	0.646	3
Pooled	0.437	1085	0.360	68

Table 10: Complete-case ECS_{adj} split by whether the model’s prediction was correct. Descriptive only: the incorrect stratum is small (68 of 1,153 complete cases) and the per-cell direction is inconsistent (agreement is higher on incorrect predictions in 4 of 11 estimable cells).

Regime (b) is the dominant one and, importantly, is *not* confined to short inputs. CAD-IMDb—our *longest* inputs, paragraph-length reviews with median $V = 68$ —tops the exclusion counts at 398 pair-slots, and every one of them is regime (b): its median H set holds 30 tokens against a median CF set of 11 and R of 6, giving a median $J_{\max}$ of 0.108 among excluded pairs. AG News is likewise 48/48 regime (b). MNLI (248 of 289) and SST-2 (148 of 213) mix the two, since short premises produce both small V and one-token counterfactuals. Totals: CAD-IMDb 398, MNLI 289, SST-2 213, AG News 48 (948, reconciling with the pair-type view above).

Because the guard falls hardest on H- and CF-involving pairs, it could in principle shape the ladder. It does not: restricting to the 850 complete cases containing *no* degenerate pair reproduces the ordering and the separations (E–P 0.618, E–R 0.451, R–P 0.313; full complete-case set 0.615, 0.395, 0.286). E–R rises the most under the restriction (+0.056), which narrows the E–P/E–R gap while leaving R–P clearly lowest.

F Disattenuated Agreement, Per Cell

Table 11 reports, for every cell and pair: observed AJ, the two reliabilities, the corrected value with bootstrap CI, at-ceiling flags, and the pre-registered floor exclusions (CF on Nova Pro/AG News, where CF paraphrase reliability is 0.10).

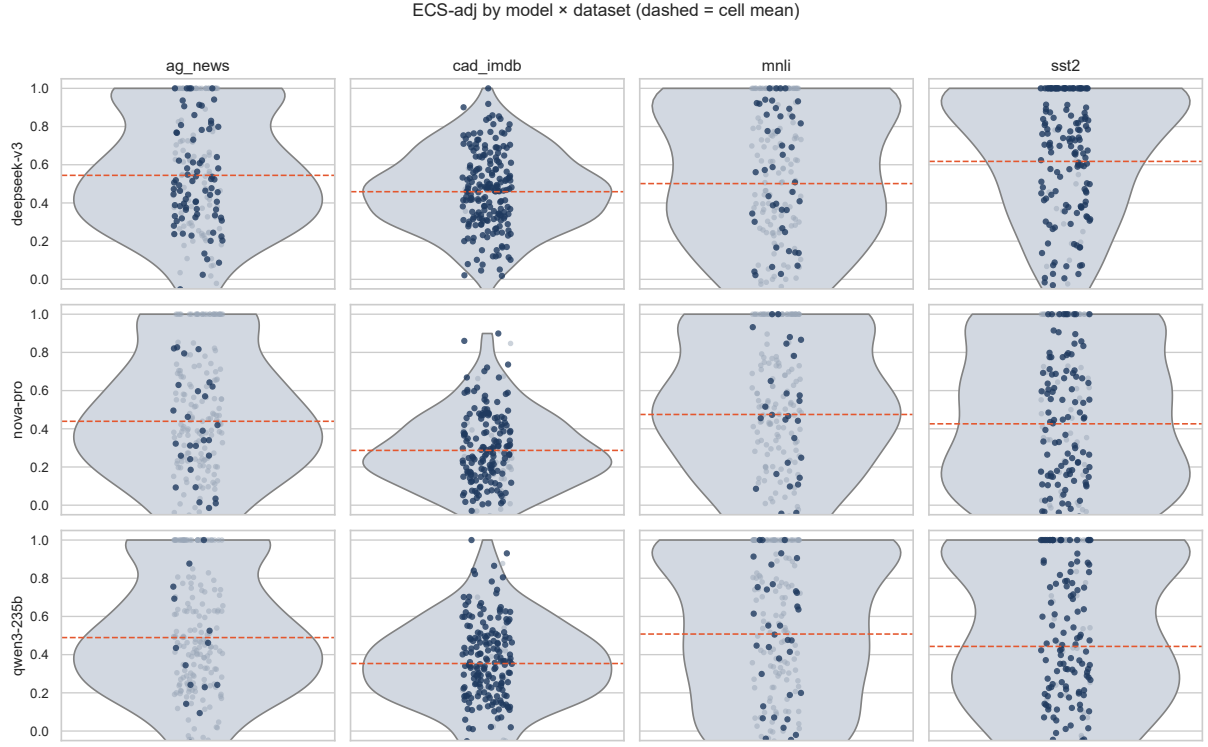


Figure 2: Per-instance ECS_{adj} distributions by model (rows) and dataset (columns); dashed line = cell mean; dark points = complete-case instances. All 12 cell means are above chance (Table 1), but distributions are wide and often bimodal.

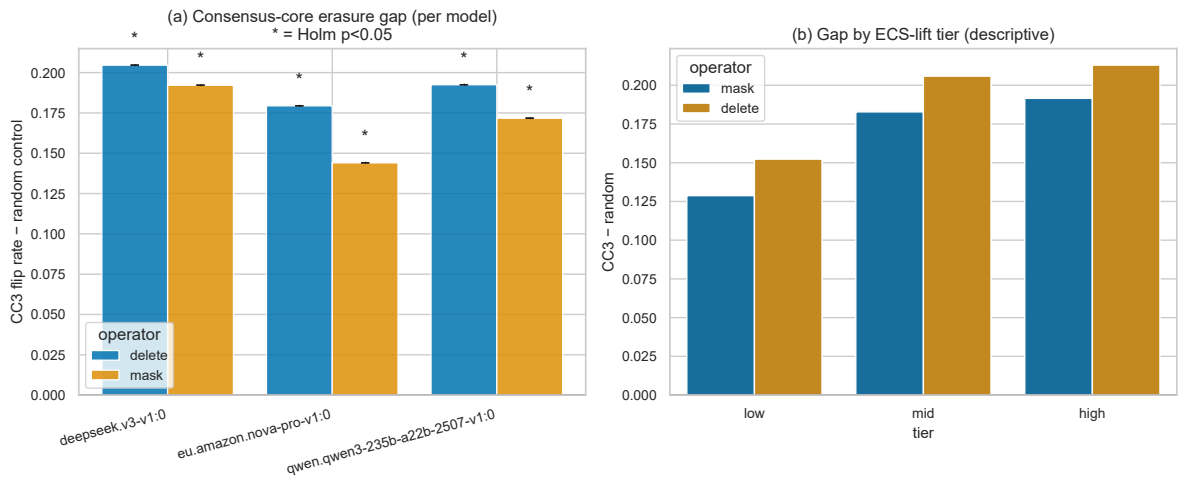


Figure 3: Consensus-core erasure. (a) CC3-minus-random flip-rate gap per model and operator (all Holm $p < .05$). (b) Pooled gap by ECS_{adj} -lift tier (descriptive): higher cross-paradigm agreement coincides with a larger causal-sensitivity gap.

Table 11: Disattenuated cross-paradigm agreement (Spearman 1904): observed AJ corrected by the geometric mean of the two strategies’ self-consistency ceilings, per cell. Pairs below the pre-registered reliability floor (0.30, ceiling $n \geq 10$) are reported as excluded.

Cell	Pair	Obs.	$rel_A \times rel_B$	Corrected [95% CI]	Flag
DeepSeek-V3/AG News	H-R	0.534	0.742×0.878	0.662 [0.585, 0.741]	at ceiling
DeepSeek-V3/AG News	RO-R	0.548	0.786×0.878	0.659 [0.587, 0.727]	
DeepSeek-V3/AG News	H-CF	0.569	0.742×0.305	1.197 [1.013, 1.428]	
DeepSeek-V3/AG News	RO-CF	0.479	0.786×0.305	0.979 [0.797, 1.192]	
DeepSeek-V3/AG News	R-CF	0.456	0.878×0.305	0.882 [0.692, 1.099]	
DeepSeek-V3/CAD-IMDb	H-R	0.631	0.588×0.759	0.945 [0.852, 1.046]	at ceiling
DeepSeek-V3/CAD-IMDb	RO-R	0.345	0.880×0.759	0.422 [0.368, 0.480]	
DeepSeek-V3/CAD-IMDb	H-CF	0.616	0.588×0.797	0.899 [0.834, 0.967]	
DeepSeek-V3/CAD-IMDb	RO-CF	0.797	0.880×0.797	0.951 [0.904, 1.000]	
DeepSeek-V3/CAD-IMDb	R-CF	0.277	0.759×0.797	0.357 [0.310, 0.406]	
DeepSeek-V3/MNLI	H-R	0.495	0.660×0.943	0.627 [0.535, 0.725]	at ceiling
DeepSeek-V3/MNLI	RO-R	0.551	0.746×0.943	0.657 [0.577, 0.740]	
DeepSeek-V3/MNLI	H-CF	0.338	0.660×0.554	0.558 [0.162, 0.937]	
DeepSeek-V3/MNLI	RO-CF	0.546	0.746×0.554	0.850 [0.624, 1.090]	
DeepSeek-V3/MNLI	R-CF	0.367	0.943×0.554	0.508 [0.259, 0.777]	
DeepSeek-V3/SST-2	H-R	0.498	0.743×0.910	0.605 [0.510, 0.709]	at ceiling
DeepSeek-V3/SST-2	RO-R	0.547	0.883×0.910	0.610 [0.522, 0.694]	
DeepSeek-V3/SST-2	H-CF	0.753	0.743×0.648	1.085 [0.967, 1.217]	
DeepSeek-V3/SST-2	RO-CF	0.778	0.883×0.648	1.028 [0.925, 1.138]	
DeepSeek-V3/SST-2	R-CF	0.581	0.910×0.648	0.756 [0.629, 0.884]	
Nova Pro/AG News	H-R	0.443	0.571×0.944	0.604 [0.520, 0.684]	excl.: rel. floor (CF)
Nova Pro/AG News	RO-R	0.450	0.652×0.944	0.573 [0.502, 0.649]	
Nova Pro/AG News	H-CF	0.379	0.571×0.096	–	
Nova Pro/AG News	RO-CF	0.332	0.652×0.096	–	
Nova Pro/AG News	R-CF	0.275	0.944×0.096	–	
Nova Pro/CAD-IMDb	H-R	0.444	0.500×0.807	0.699 [0.591, 0.809]	at ceiling
Nova Pro/CAD-IMDb	RO-R	0.201	0.712×0.807	0.265 [0.212, 0.321]	
Nova Pro/CAD-IMDb	H-CF	0.318	0.500×0.710	0.534 [0.466, 0.604]	
Nova Pro/CAD-IMDb	RO-CF	0.740	0.712×0.710	1.041 [0.974, 1.114]	
Nova Pro/CAD-IMDb	R-CF	0.090	0.807×0.710	0.119 [0.065, 0.170]	
Nova Pro/MNLI	H-R	0.529	0.625×0.928	0.695 [0.595, 0.800]	at ceiling
Nova Pro/MNLI	RO-R	0.536	0.757×0.928	0.640 [0.557, 0.726]	
Nova Pro/MNLI	H-CF	0.330	0.625×0.446	0.626 [0.212, 1.039]	
Nova Pro/MNLI	RO-CF	0.171	0.757×0.446	0.295 [0.001, 0.597]	
Nova Pro/MNLI	R-CF	0.191	0.928×0.446	0.297 [-0.027, 0.632]	
Nova Pro/SST-2	H-R	0.466	0.541×0.884	0.674 [0.554, 0.795]	at ceiling
Nova Pro/SST-2	RO-R	0.415	0.735×0.884	0.515 [0.410, 0.626]	
Nova Pro/SST-2	H-CF	0.431	0.541×0.648	0.727 [0.571, 0.884]	
Nova Pro/SST-2	RO-CF	0.541	0.735×0.648	0.784 [0.662, 0.912]	
Nova Pro/SST-2	R-CF	0.331	0.884×0.648	0.437 [0.300, 0.573]	
Qwen3-235B/AG News	H-R	0.487	0.731×0.936	0.589 [0.518, 0.661]	at ceiling
Qwen3-235B/AG News	RO-R	0.480	0.817×0.936	0.549 [0.493, 0.610]	
Qwen3-235B/AG News	H-CF	0.465	0.731×0.620	0.691 [0.353, 1.030]	
Qwen3-235B/AG News	RO-CF	0.434	0.817×0.620	0.611 [0.281, 0.946]	
Qwen3-235B/AG News	R-CF	0.410	0.936×0.620	0.538 [0.179, 0.906]	
Qwen3-235B/CAD-IMDb	H-R	0.622	0.645×0.863	0.834 [0.735, 0.923]	at ceiling
Qwen3-235B/CAD-IMDb	RO-R	0.163	0.876×0.863	0.187 [0.144, 0.230]	
Qwen3-235B/CAD-IMDb	H-CF	0.647	0.645×0.806	0.897 [0.829, 0.971]	
Qwen3-235B/CAD-IMDb	RO-CF	0.752	0.876×0.806	0.895 [0.839, 0.955]	
Qwen3-235B/CAD-IMDb	R-CF	0.138	0.863×0.806	0.166 [0.119, 0.216]	
Qwen3-235B/MNLI	H-R	0.549	0.743×0.975	0.645 [0.561, 0.727]	at ceiling
Qwen3-235B/MNLI	RO-R	0.533	0.800×0.975	0.604 [0.528, 0.686]	
Qwen3-235B/MNLI	H-CF	0.506	0.743×0.655	0.725 [0.407, 1.020]	
Qwen3-235B/MNLI	RO-CF	0.427	0.800×0.655	0.590 [0.371, 0.795]	
Qwen3-235B/MNLI	R-CF	0.338	0.975×0.655	0.424 [0.145, 0.711]	
Qwen3-235B/SST-2	H-R	0.319	0.721×0.924	0.390 [0.280, 0.496]	at ceiling
Qwen3-235B/SST-2	RO-R	0.359	0.853×0.924	0.404 [0.311, 0.502]	
Qwen3-235B/SST-2	H-CF	0.744	0.721×0.822	0.967 [0.864, 1.065]	
Qwen3-235B/SST-2	RO-CF	0.751	0.853×0.822	0.896 [0.809, 0.989]	
Qwen3-235B/SST-2	R-CF	0.258	0.924×0.822	0.296 [0.174, 0.425]	

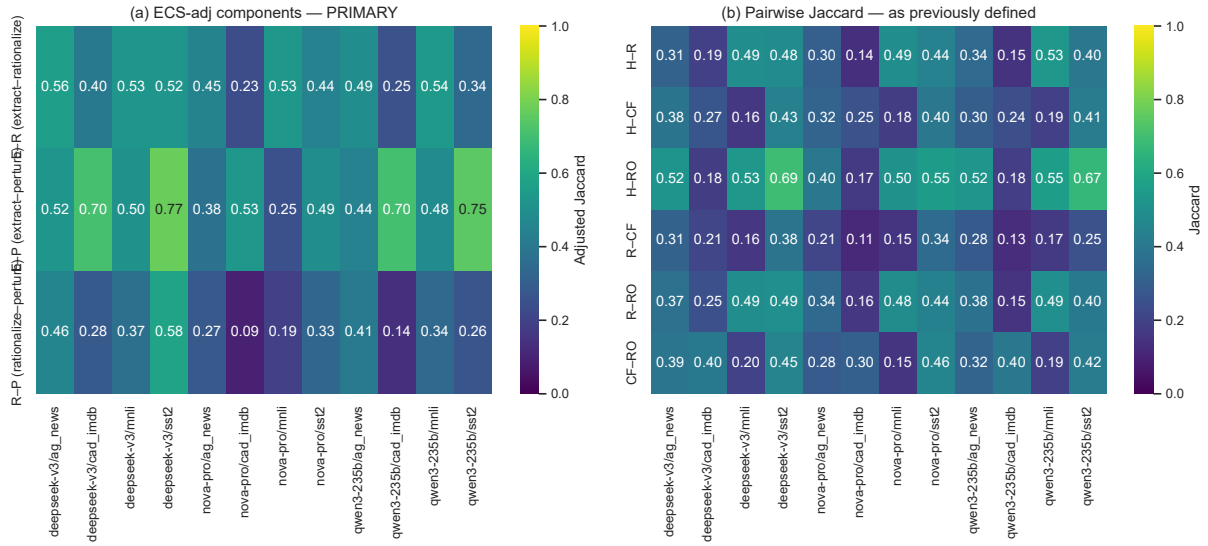


Figure 4: (a) ECS_{adj} paradigm components per cell (primary scale). (b) The raw pairwise Jaccard view of the same data (deprecated comparator): without chance/ceiling adjustment, set-size geometry dominates the picture.

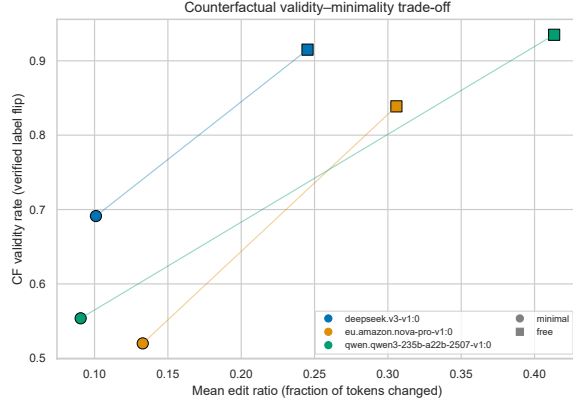


Figure 5: Counterfactual validity–minimality trade-off: minimal-edit CFs flip less often but edit far less; free rewrites flip reliably (90%) but over-edit.

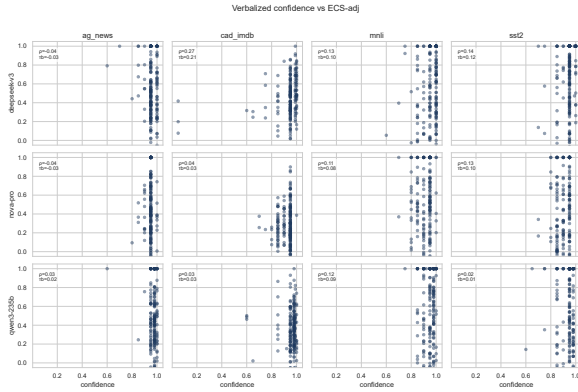


Figure 6: Verbalized confidence vs. ECS_{adj} per cell. Confidence is degenerate at temperature 0 (clustered at 0.95), consistent with Xiong et al. (2024); associations are weak and inconsistent.